



ITSimplera Institute

WEEK 01: Configuring a Simple LAN network using Packet tracer and GNS3

Domain: Network administration

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Simple LAN Network Implementation Report

Cisco Packet Tracer and GNS3 – Week 1 Task

1. Introduction

This report documents the design, configuration, and testing of a simple Local Area Network (LAN) built using two different network simulation tools: Cisco Packet Tracer and GNS3. The purpose of this task was to gain practical, hands-on experience with basic networking concepts by building an identical network topology in both environments, comparing the two tools, and verifying connectivity between the connected devices.

The network consisted of four computers connected in a star topology through a central switch, with each device assigned a unique static IP address. Once the network was built, ping tests were carried out between the devices to confirm that communication was successful across the LAN. The complete task, including screenshots, configuration steps, and results, was organized into two separate project folders, one for each tool, along with a supporting PDF report.

Networking Concepts Learned

This exercise reinforced several foundational networking concepts:

Star Topology: All four PCs were connected individually to a central switch rather than to each other directly.

This design means every device has its own dedicated link to the switch, so if one cable or PC fails, the rest of the network keeps working normally.

Switching: The switch operates at Layer 2 and forwards traffic between connected devices based on MAC addresses, allowing all four PCs to communicate with one another without needing a router, since they belong to the same local network.

IP Addressing: Each PC was manually configured with a static IPv4 address in the range 195.1.1.1 to 195.1.1.4, all within the same subnet, which is what allows the devices to reach each other directly without additional routing.

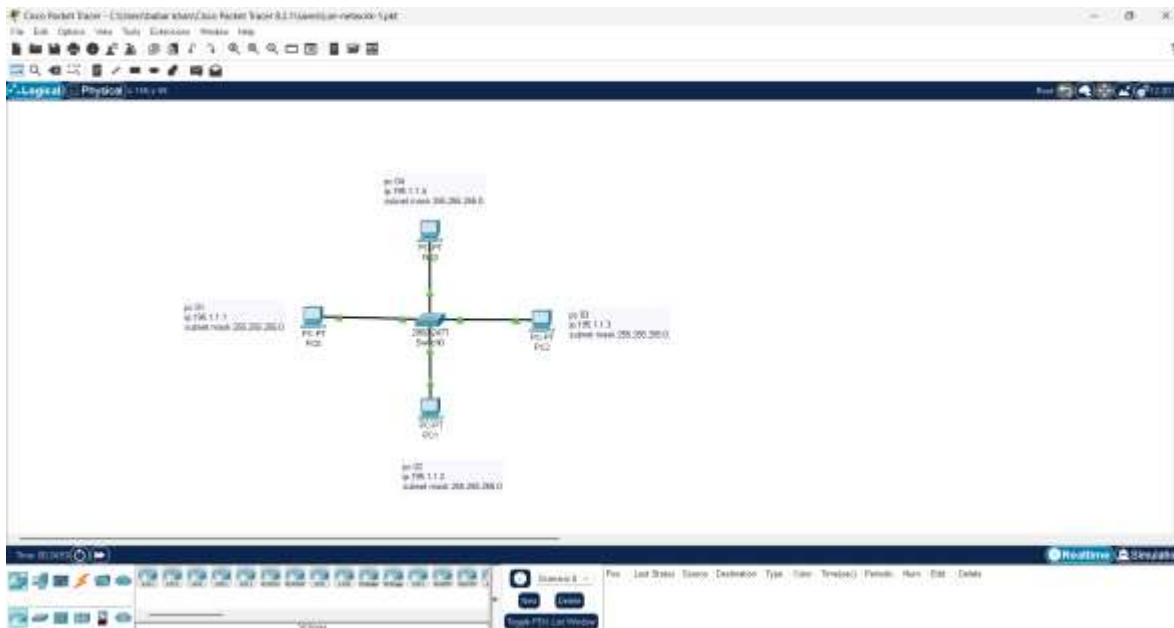
Connectivity Testing: The ping utility (ICMP Echo Request/Reply) was used to verify that each device could successfully reach the others, confirming that both the physical/logical connections and the IP configuration were correct.

Simulation vs. Emulation: Packet Tracer simulates network behaviour at a conceptual level and is aimed at learning, while GNS3 emulates real device operating systems, giving a closer-to-real-world experience of configuring network equipment.

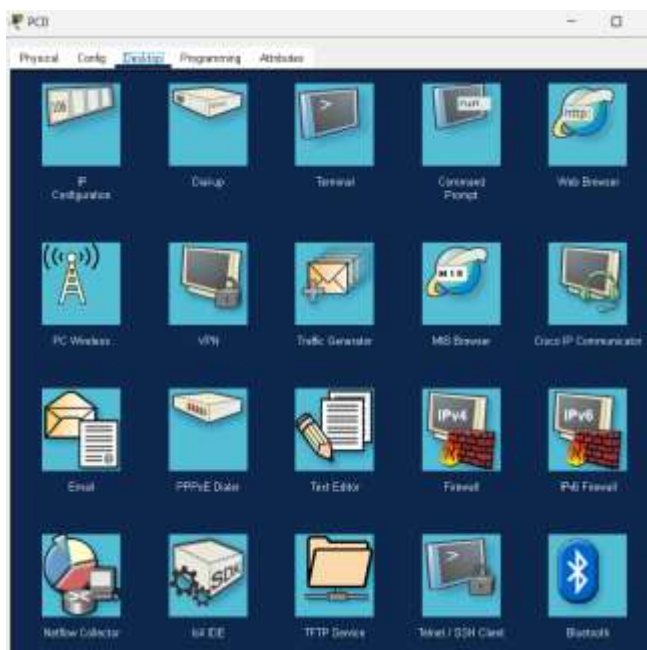
Cisco Packet Tracer Interface Overview:

Cisco Packet Tracer provided a straightforward, beginner-friendly interface. Devices such as PCs and switches could be dragged directly from the device category panel on the bottom-left into the workspace. Connections between devices were made using the appropriate cable type from the connections menu, and clicking on a device opened a configuration window where the IP address and subnet mask could be entered under the Desktop tab's IP Configuration utility. The simulation mode also allowed packets to be tracked visually as they moved across the topology, which made it easy to confirm that the star topology was wired correctly before running any ping tests.

Packet tracer interface & LAN TOPOLOGY:

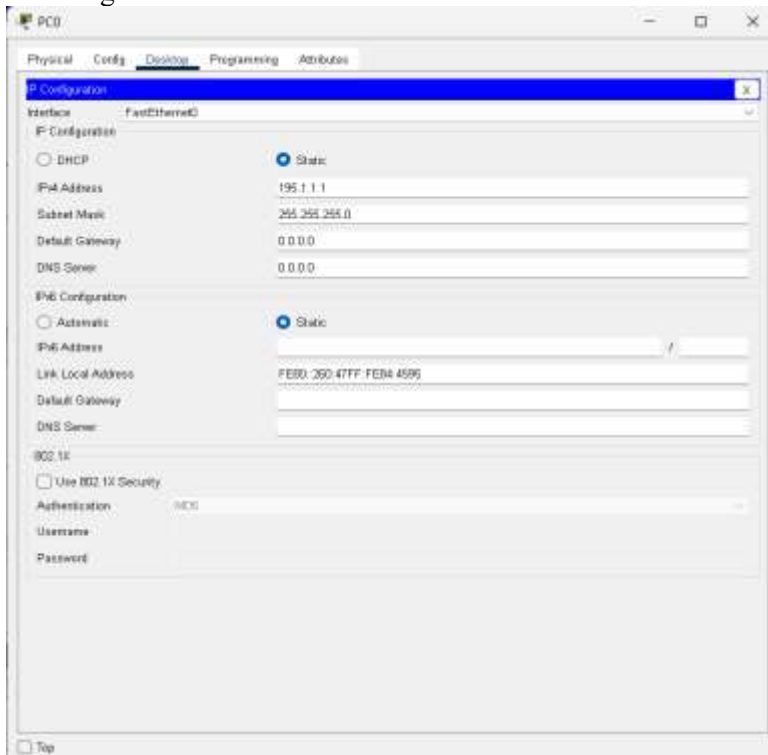


Right clicking a pc opens this interface:



IP configuration of PC:

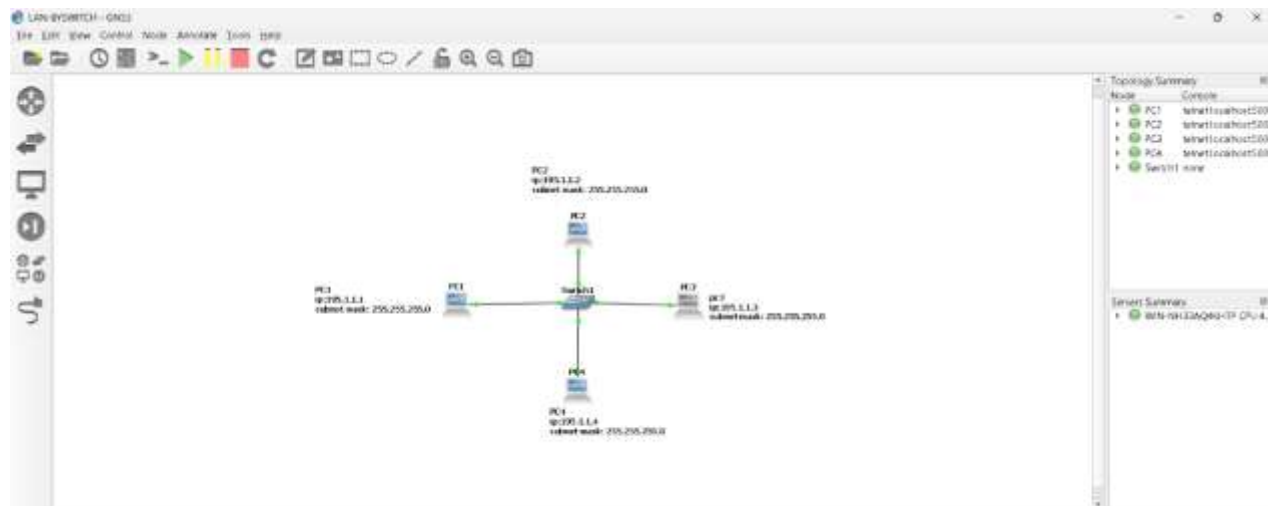
IP configuration window



GNS3 Interface Overview:

GNS3 offered a more advanced, professional-style interface intended for closer-to-real device emulation. Devices were added from a components panel and linked together using virtual cables, similar in principle to Packet Tracer, but with a steeper learning curve. IP addresses on each PC node were configured through the node's console or settings, and the overall workspace required more manual setup, such as starting device nodes individually before they became active on the topology. Compared to Packet Tracer, GNS3 felt closer to working with real network hardware and software.

GNS3 interface & LAN topology:



Left clicking a pc and selecting console opens this PUTTY interface that acts as a command prompt

IP Configuration of PC:

```

PC1 - PuTTY
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All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ping 195.1.1.3
host (195.1.1.3) not reachable

PC1> ip 195.1.1.1 255.255.255.0
Checking for duplicate address...
PC1 : 195.1.1.1 255.255.255.0

PC1> save
Saving startup configuration to startup.vpc
. done

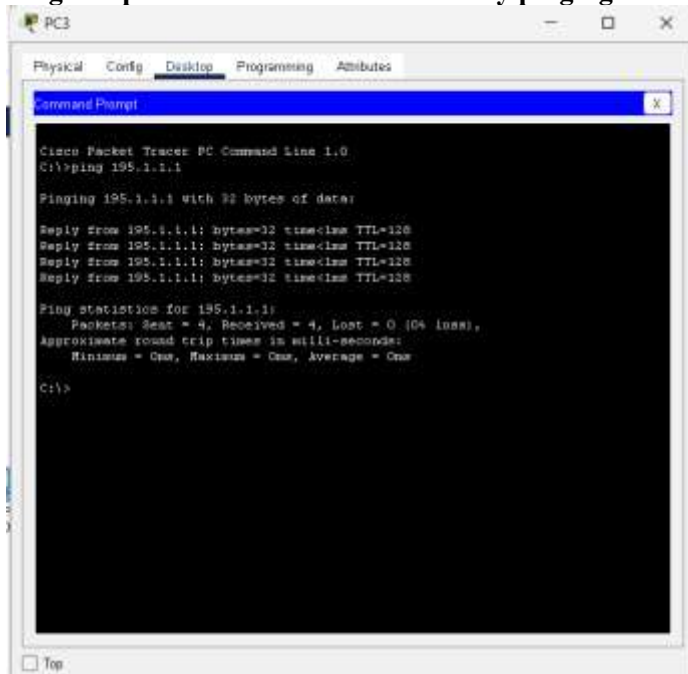
PC1>

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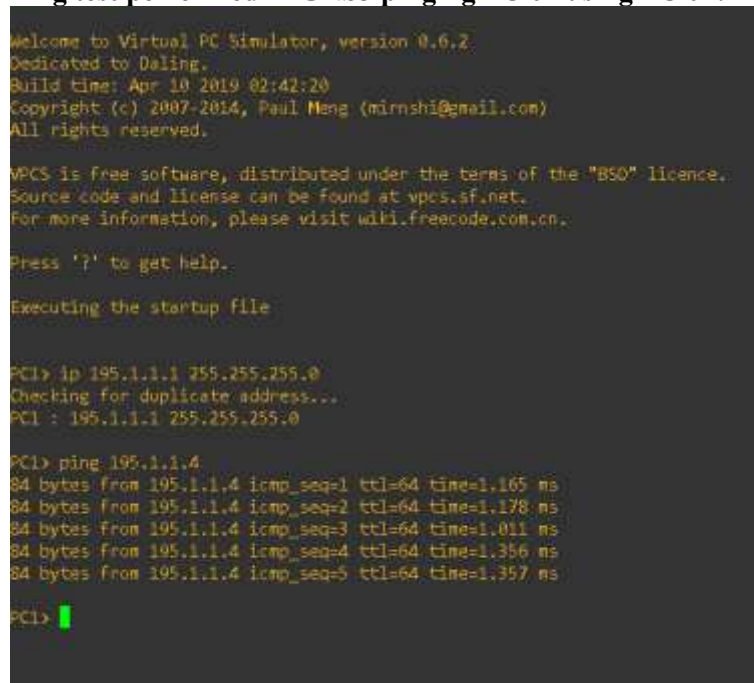
Ping Test Results:

After configuring all four PCs with their static IP addresses and connecting them to the switch in a star topology, ping tests were performed between multiple pairs of devices to confirm end-to-end connectivity across the LAN. All tests returned successful replies with no packet loss, confirming that the IP addressing scheme and physical/logical topology were configured correctly in both Packet Tracer and GNS3.

Ping test performed in Packet Tracer by pinging PC 01 using PC 03:



Ping test performed in GNS3 pinging PC 04 using PC 01:



Challenges Faced

The main challenge encountered during this task was navigating the GNS3 user interface, since it was my first time using the application. Unlike Packet Tracer, which has a simplified drag-and-drop workflow suited for beginners, GNS3's interface required more familiarity with networking tools and concepts. Locating the correct device templates, understanding how to link nodes properly, and starting/configuring the PC nodes took extra

time and some trial and error before the topology matched the one built in Packet Tracer. Once the interface became more familiar, the remaining steps, cabling, IP configuration, and ping testing, were completed without further issues.

2. Task Organization and Deliverables:

The completed work for this task has been organized into the following files and folders for submission:

GNS3 Week 1 – a folder containing the GNS3 project files and screenshots showing the star topology, switch connections, and PC configurations built in GNS3.

Packet Tracer Week 1 – a folder containing the Packet Tracer project file (.pkt) and screenshots showing the equivalent star topology, switch connections, and PC configurations built in Packet Tracer.

Task Report (PDF) – a separate PDF file containing a detailed report of the task, including step-by-step configuration screenshots and the ping test outputs from both tools.

Keeping the two tools in separate folders made it easier to compare the workflow and results of each platform side by side, while the accompanying PDF captures the visual evidence (screenshots and command outputs) that supports the summary provided in this report.

3. Conclusion

This task provided practical experience in building a simple LAN from the ground up, from planning a star topology around a central switch, to assigning static IP addresses, to verifying connectivity with ping tests. Completing the exercise in both Cisco Packet Tracer and GNS3 highlighted the differences between a simplified learning-focused simulator and a more realistic emulation platform. Although GNS3's interface presented an initial learning curve, working through it strengthened both my technical understanding of LAN design and my confidence in using industry-style networking tools. The successful ping results between all four PCs (195.1.1.1 through 195.1.1.4) confirm that the network was configured correctly in both environments, meeting the objectives of this task.